TFLBot: A Helping Hand Around The Station

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***Abstract*—Navigating large metropolitan public transport systems can be challenging, particularly for first-time users and passengers with accessibility needs. Complex station layouts, static signage, and reliance on staff availability can increase cognitive load and travel-related anxiety. This project explores the feasibility of TFLBot, a mobile assistive robot designed to support passengers in underground transport environments through autonomous navigation guidance and accessible, multimodal interaction. The system combines autonomous mobility with speech, visual feedback, and British Sign Language (BSL) to provide informative and inclusive assistance. This report presents the system design, implementation decisions, and evaluation approach for assessing the effectiveness of robotic guidance in complex public transport settings.**

I. Introduction

This project investigates the effectiveness of a mobile assistive robot for passenger guidance in underground transport environments. Specifically, it evaluates two assistance modes: (i) a mobile, embodied guidance mode, in which the robot physically escorts users and communicates through verbal and non-verbal cues, and (ii) a stationary, instruction-based mode, in which guidance is provided from a fixed location.

The TFLBot is designed to operate autonomously within metropolitan stations. Users approach the robot and ask for help with navigation or information. The system then integrates autonomous mobility, perception, and multimodal interaction to guide users or answer questions. TFLBot is designed to provide guidance using both modes for testing.

One important focus is the use of non-verbal and embodied communication to reduce cognitive demands associated with navigation. By shifting the burden from interpreting verbal instructions or maps, the system aims to improve navigation efficiency while lowering user cognitive load.

This report presents the design and implementation of the TFLBot system and describes an experimental methodology that compares the two interaction modes. Performance and user experience are evaluated using objective navigation metrics and subjective workload and satisfaction measures, with the aim of assessing the viability of robotic guidance in underground public transport settings.

II. Hypothesis

When navigating a station, a mobile assistive robot that provides physical guidance with verbal and visual cues will offer an improved user experience and guidance performance compared to a stationary robot that only provides verbal directions.

Following initial research outlined in Section III, it was theorised that incorporating Pepper's mobility and hand-gesturing capabilities into a train station guide robot could significantly enhance user interaction. Unlike a stationary system limited to verbal instruction, a mobile assistant is able to physically accompany users through the station, providing real-time guidance while reinforcing directions through gestures and visual indications.

This approach is expected to improve clarity, accessibility, and user confidence, particularly in complex or crowded stations, where verbal instructions alone may not be sufficient.

III. Background Research

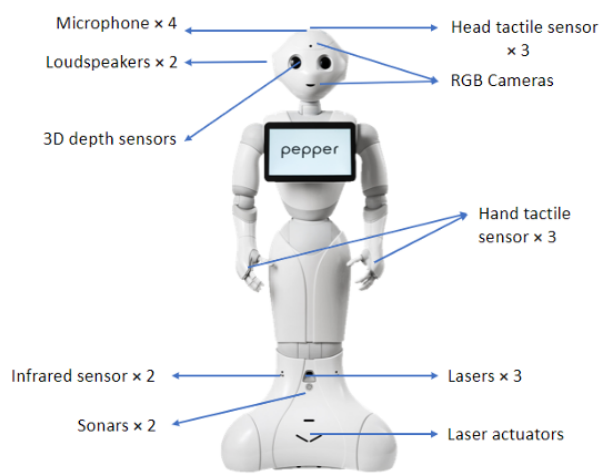

Fig. 1. Pepper Bot

A. Softbank Pepper

The Pepper robot [1] is a 1.2-meter-tall social humanoid developed by SoftBank Robotics in 2014, engineered with 17 joints for fluid movement and a 12-hour battery life to serve in education, healthcare, and hospitality. Its hardware architecture features a sophisticated sensory suite including 4 microphones, 2 loudspeakers, 3D depth sensors, and lasers.

Empirical research, however, reveals significant technical inconsistencies. Specifically, the built-in SLAM navigation is considered unreliable due to a reported average localisation error of 0.51 meters and a blocking mapping process that prevents the robot from executing other commands until exploration is complete. While Pepper demonstrates satisfactory

sound localisation and speech recognition, its facial recognition efficiency drops significantly beyond a range of 2 meters, and the robot often experiences confusion when managing multiple simultaneous speech inputs or navigating in extreme lighting conditions.

In addition to this, Pepper's onboard processing, based on an 1.9 GHz quad-core Atom processor with approximately 2 GB of usable RAM and no dedicated GPU, offers limited processing power for modern vision and perception tasks, which commonly require multi-core CPUs and 8–32 GB RAM.

B. Orbbec Femto Bolt

Upgrading the Pepper robot with an Orbbec Femto Bolt Camera [2] provides a significant leap in spatial awareness and environmental interaction by replacing the original, often unreliable depth system with 1 megapixel Time-of-Flight (ToF) technology. This camera offers a high-resolution 1024x1024 depth map and a wide 120-degree field of view, which directly addresses Pepper's known navigation flaws, such as the 0.51-meter localisation error and poor obstacle detection in complex lighting. Furthermore, the Femto Bolt integrates a 4K HDR RGB camera and a 6-axis IMU, enabling precise depth-to-colour registration and better orientation tracking. Because it is compatible with the Microsoft Azure Kinect SDK and features a built-in Human Skeleton Tracking SDK, it enables the robot to track bodies and recognise gestures with much higher reliability than the stock hardware, especially beyond the typical 2-meter visual range where the original sensors often fail.

C. Robotic Assistance in Public Transport Environments

Robotic systems have been deployed in public transport environments to support passenger assistance, primarily through information provision and navigation support. One example is the Pepper robot, which has been trialled in metro stations and railway terminals to provide directions and service information via speech and an integrated display [3]. These deployments typically position the robot as an information point rather than an active guide.

More recently, commercial platforms such as the LG Airport Guide Robot and LG CLOi GuideBot have been introduced in large transport hubs, where they provide mobile guidance by escorting users to destinations while delivering navigational information [4], [5]. As combined guides and information points, they offer useful reference points for improving user experience, including features such as multilingual support.

D. Mobile Escort and Wayfinding Robots

Beyond transport-specific deployments, a broader class of mobile robots has been developed for escort and wayfinding tasks in complex public environments. The SPENCER robot demonstrated that embodied guidance can reduce navigation ambiguity by allowing users to follow the robot directly rather than interpret verbal directions [6].

Commercial platforms such as the temi personal robot have also been adapted for escort-style guidance in airports and large venues [7]. However, these systems are typically designed for open environments and provide limited accessibility-focused interaction.

E. Social and Concierge-Oriented Service Robots

Another class of robotic systems prioritises social engagement, with navigation treated as a secondary function. Robots such as Mirokaï emphasise expressive motion and conversational interaction to improve approachability in public spaces [8]. Similarly, Pepper is frequently deployed in concierge roles rather than as an active guide [9]. Though in these systems, mobility primarily enhances presence rather than navigation capability, they provide a useful reference for designing and improving communication.

F. Accessibility and Interaction Limitations

Existing systems remain largely focused on speech- and screen-based interaction. While embodied guidance through physical motion has been shown to reduce navigation ambiguity, non-verbal communication cues such as gesture are often limited or treated as secondary interaction elements [6], [10]. Support for accessibility-focused interaction, including sign language or adaptive multimodal feedback, remains rare in deployed transport robots.

Furthermore, evaluations of robot guides often prioritise technical performance and navigation success over user-centred measures such as cognitive load or satisfaction, limiting insight into the benefits of different guidance modes.

IV. System Design

The overall system design comprises of multiple subcomponents (as seen in Figure 2) as a result of the Pepper's limited sensory and onboard processing capabilities, providing greater flexibility in the control architecture, as each module can be more directly interfaced with compared to using the Pepper itself. Note the inclusion of the Raspberry Pi 5 [11] and a Jetson Orin Nano [12] to allow for multiple simultaneous commands and image processing of data from the Femto Bolt Camera. We also make use of Gemini [13] to process speech received from the microphone, and also real time TFL Status updates. All of these additional components are to be powered by the Anker Solix C300 DC battery [14].

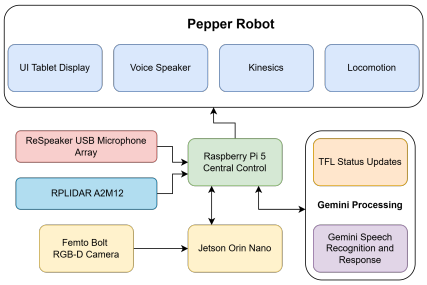

Fig. 2. System Design Diagram

A. Speech Recognition

To facilitate user–robot interaction in the Underground, the robot incorporates a speech recognition submodule which allows it to recognise spoken input and initiate responses. The speech recognition system is implemented using Python's SpeechRecognition library, which provides a method for converting spoken audio into text [15]. This library supports multiple recognition engines and also includes built-in noise reduction capabilities in order to improve the accuracy in real-world environments.

The recognised speech commands are processed through Gemini AI [13], allowing the robot to provide current route information and respond intelligently to complex navigation queries. Text-to-speech conversion is then used to output Gemini's [13] response, which will be handled by the **pyttsx3** library, enabling the robot to deliver spoken responses to users [15].

To evaluate whether a more featured robot improves user experience, the system incorporates a microphone array [16] for speaker localisation. This enables the robot to automatically turn and face users during interaction. This directional capability supports the mobile assistance features required for testing the hypothesis.

A stretch goal for speech recognition would be the inclusion of multilingual communication, allowing users to speak and hear different languages with Pepper [17].

B. Human Detection and Sign Language Recognition

This submodule enables the robot to detect nearby users and interpret sign language gestures for communication. The initial stage of this system involves human detection. An array microphone provides rough localisation data to compensate for the Femto Bolt Depth and the RGB camera's lack of a 360° field of view (FOV). Once a direction is established, the camera localises the individual to refine the interaction.

The depth camera will be used to detect the presence of an object within the robot's FOV and to determine its distance. If the person is too close, the robot will tell the user to move back or will move away from the user. Conversely, if a person exceeds a specified threshold, the system suspends computer vision processing because interaction intent is unlikely, to prevent unnecessary processing of background noise.

The current version of the program uses Google's MediaPipe to identify the person within the frame. [18] The program identifies key landmarks: face, shoulders, arms, and hands, which are all essential for British Sign Language (BSL). Figure 3 shows both camera streams of the depth camera and RGB behind. The depth stream contains a rectangular region at the centre that displays the window within which a person must be positioned for recognition. This was added because the more centred a potential user is, the more likely they are to be a user rather than a bystander. [19] A current limitation is that the system prioritises the closest individual, regardless of intent. If a passerby moves between the robot and the user, the system erroneously tracks the individual who is closer to the robot. Future protocols must be implemented to prioritise users based on both optimal distance and proximity to the centre of the FOV. Following localisation, the system treats arm and hand movements as extensions of the body to infer sign language meaning via machine learning.

Additionally, the Femto Bolt and Mediapipe have been tested in a laboratory environment, performance outdoors in both sunlight and low light must be tested.

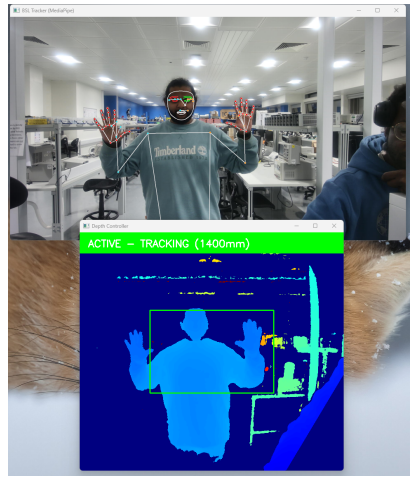

Fig. 3. Screenshot of the depth and RGB camera streams during human recognition

C. Kinesics: Arm and Upper Body Control

As part of our goal to assist travellers, we aim to design the robot's communication system to be as inclusive and informative as possible. As a part of this, we will use Pepper's head, arms, and upper body motions to transmit non-verbal communication cues to complement verbal interaction.

As a first step to allow us to integrate with the whole system, we focus on implementing a small set of simple, clearly distinguishable hand gestures, such as pointing at or showing objects or directions. These gestures will act as a base on which we can build more expressive communication. With this integration, we plan to extend the gesture set to include a limited number of words in British Sign Language. This extension will act as a proof-of-concept to evaluate the feasibility of incorporating sign language into human–robot interaction.

To enable this functionality, we first implemented remote teleoperation of Pepper's arms using Python scripts on a computer. These scripts leverage a verified open source library for the forward and inverse kinematics of the Pepper arms, `pepper kinematics` [20], to translate between joint angles and physical Cartesian coordinates in both directions. We also made use of the AlMotion service to be able to write specific joint angles and the time they should occur. This allowed us to achieve full control over the robot's arm movements, which gave us a reliable platform for gesture development and testing.

With low-level arm control established, the next step involves collaboration with the speech team to implement ges-

ture selection. In this, we will map specific words, phrases or interaction states to defined arm movement paths (gestures). For example, during a “thinking” state, Pepper’s arms may perform a dedicated gesture to visually indicate that the robot is processing information.

The final stage of this design will involve combining spoken output with corresponding simple BSL gestures, allowing Pepper to reinforce verbal communication through sign language. This aims to improve accessibility and comprehension for users with hearing impairments, while also improving overall interaction clarity.

D. User Interface

To provide intuitive human–robot interaction, we designed and implemented a user interface which includes multiple accessibility and usability features to enhance interaction. Language flags enable users to select their preferred language for communication. Real-time speech transcription displays what the robot heard, allowing users to identify and correct recognition errors. Text-based output of Gemini’s [13] responses ensures accessibility for deaf and hard-of-hearing users, as well as those in noisy station environments. Finally, a visual ”thinking” indicator provides feedback during processing delays, reassuring users that their request is being handled.

Fig. 4. Simple UI for Language selection

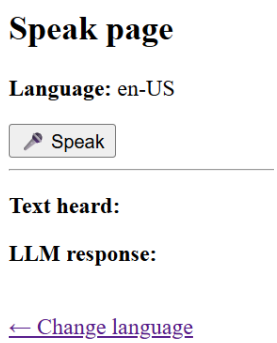

Fig. 5. UI allows the user to see what has been said

E. Path Planning and Collision Avoidance

The path planning and collision avoidance module is intended to enable the robot to autonomously navigate from an initial position to a specified goal while minimising travel distance. The overall navigation strategy is to be based on a global path-planning approach using the A* algorithm [21]. Establishing reliable collision avoidance at this stage provides a safety-critical foundation upon which the global A* planner will be integrated, allowing for efficient route planning while maintaining robustness in the presence of dynamic or partially known obstacles.

Current progress has focused on the development of a local collision avoidance system intended to act as its aid. Pepper’s infrared, sonar, and laser sensors have been tested to detect nearby obstacles, but have proved to be quite inaccurate and to only sample at infrequent intervals. For example, the plot in Figure 6 shows the data collected from the laser scanner in the Pepper robot, which was provided with some short-distance obstacles. As seen, the laser has a low sample rate, and was unable to clearly differentiate between obstacles. Going forward, a more sophisticated sensor, such as the RPLIDAR A2M12 [22] will be used, as it is more accurate and has a longer range of detection (12m).

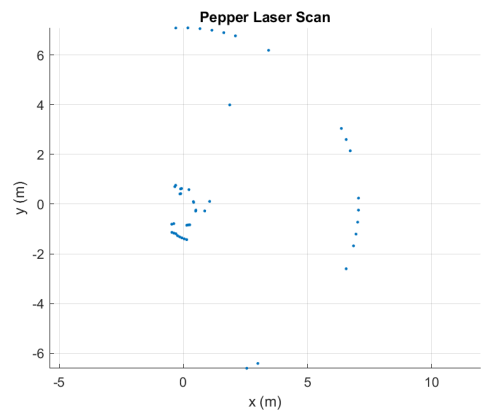

Fig. 6. Laser scan from Pepper (centred at (0,0)) with short-distance obstacles

F. Mobility and Motor Control

This functionality is separated into its own submodule to keep safety-critical collision avoidance responsive, even when global route planning is incomplete or unavailable. Integrating the naoqi driver2 package [23] enables the Pepper robot to perform advanced teleoperation and autonomous navigation within the complex environment of a tube station. By bridging the robot’s internal Naoqi OS with the ROS 2 Humble framework, the system can leverage professional-grade navigation stacks to move beyond simple manual control. The robot utilises a pre-loaded occupancy grid map of the station to plan efficient paths while simultaneously processing real-time data from its onboard lasers, sonars, and the Femto Bolt camera to detect and avoid dynamic obstacles like commuters. This combination of global mapping and local sensory feedback ensures that the robot can maintain its trajectory through narrow corridors and busy platforms, providing a reliable platform for mobile assistance or facility monitoring.

Future implementations will include using quadratic interpolation in the robot’s velocity and an instant stop to ensure safety between the robot and humans walking.

Simultaneously, the safety architecture will be upgraded from the current polling-based cheque to an interrupt-driven instant stop method to assign participants to each system to

minimise bias, randomly monitor the laser and sonar sensors at a frequency, ensuring that performance differences do not arise from the safety threshold ($d < 0.5m$). This thread will bypass the standard input buffer and inject a zero-velocity command directly into the ALMotion service, ensuring immediate deceleration regardless of the user's current input or network latency.

V. Experimental Methodology

To assess the impact of mobility and enhanced communication abilities on user experience, we will conduct a comparative user study in a controlled underground-station-like environment. In our case, we will use the Electrical and Electronic Engineering building as a safe, controlled proxy with classrooms acting as platforms. Users will be asked to navigate predefined routes, such as moving from the Ground Floor Lobby to room 406 on the 4th Floor, with assistance from one of the following systems:

1) a mobile robot which physically guides you to the room, providing visual indicators for support
2) a stationary robot providing verbal directions only

The results for the same routes will be compared using the following metrics:

Performance metrics

- *Task completion time:* time taken to reach the target destination
- *Navigation errors:* number of wrong turns or backtracking events

User experience metrics

- *User cognitive load:* measured using post-task NASA Task Load Index survey [24]
- *User satisfaction and perceived usefulness:* collected via Likert-scale survey questions

We plan to randomly assign participants to each system to minimise bias. Both systems will provide equivalent information to ensure that performance differences arise solely from the type of interaction. External factors such as crowding and time of day will be controlled during the study to minimise influence on metrics such as cognitive load and user satisfaction.

A statistically significant improvement in either metric group for the robot with mobile and expressive features would support our hypothesis.

VI. Conclusion

This design report presents the system architecture and plans for evaluation methodology for the TFLBot, a mobile robotic guide for navigating public transport networks. The system will integrate a SoftBank Pepper robot with enhanced computation and sensing, including a Raspberry Pi 5, an NVIDIA Jetson Orin Nano, an Orbbec Femto Bolt camera, a microphone array, and an RPLIDAR A2M12. This report also details our research hypothesis and a review of relevant background literature that informed the system design and component selection.

References

[1] D. Mishra, G. A. Romero, A. Pande, B. N. Bhuthegowda, D. Chaskopoulos, and B. Shrestha, "An exploration of the pepper robot's capabilities: Unveiling its potential," Norwegian University of Science and Technology (NTNU), Tech. Rep., 2024.

[2] O. D. T. International, "Orbbec femto bolt hardware specifications and robotics applications," Orbbec, Tech. Rep., 2024. [Online]. Available: https://www.orbbec.com/products/tof-camera/femto-bolt/

[3] G. Robotics, "Pepper robot deployed in train stations by sncf," https://www.gwsrobotics.com/robots-airports-stations, 2015, pepper was deployed in three train stations for passenger information tasks between December 2015 and March 2016.

[4] LG Electronics, "Lg introduces airport guide robot," https://www.lgcorp.com/media/release/7871, 2017.

[5] ——, "Lg cloi guidebot: Indoor service robot," https://www.lg.com/global/business/service-robots, 2019.

[6] K. O. Arras, D. Cerqui *et al.*, "Spencer: A socially aware service robot for passenger guidance and help in busy airports," *IEEE Transactions on Intelligent Transportation Systems*, vol. 19, no. 8, pp. 2541–2553, 2018.

[7] Roboteam Home, "temi: The personal robot," https://www.robotemi.com, 2020.

[8] Enchanted Tools, "Mirokaï: A social robot designed for public spaces," https://enchanted.tools/robot, 2023.

[9] SoftBank Robotics Europe, "Pepper as a service and concierge robot," https://www.softbankrobotics.com/emea/en/pepper, 2019.

[10] N. J. Hetherington and E. A. Croft, "Nonverbal motion legibility cues for human–robot spatial interaction," *arXiv preprint arXiv:2104.02275*, 2021.

[11] *Raspberry Pi Documentation*, Raspberry Pi Foundation, 2024. [Online]. Available: https://www.raspberrypi.com/documentation/

[12] *Jetson Orin Nano Developer Kit User Guide*, NVIDIA, 2023. [Online]. Available: https://developer.nvidia.com/embedded/learn/jetson-orin-nano-devkit-user-guide

[13] Google, "Gemini api documentation," 2024, accessed: 28-Jan-2026. [Online]. Available: https://ai.google.dev/gemini-api/docs

[14] *User Guide for :contentReference[oaicite:0]index=0*, Anker SOLIX, 2024. [Online]. Available: https://support.ankersolix.com/s/article/Anker-SOLIX-C300-Portable-Power-Station-User-Guide-A1722

[15] GeeksforGeeks, "Python: Convert speech to text and text to speech," https://www.geeksforgeeks.org/python/python-convert-speech-to-text-and-text-to-speech/, 2026, accessed: 28-Jan-2026.

[16] Seeed Studio, "Respeaker usb mic array," https://wiki.seeedstudio.com/ReSpeaker-USB-Mic-Array/, 2024, accessed: 28-Jan-2026.

[17] SoftBank Robotics, "Pepper: The humanoid and programmable robot," https://www.softbankrobotics.com/emea/en/pepper, 2024, accessed: 28-Jan-2026.

[18] *MediaPipe Solutions guide*, Google. [Online]. Available: ttps://ai.google.dev/edge/mediapipe/solutions/guide

[19] M. P. Michalowski, S. Sabanovic, and R. Simmons, *A Spatial Model of Engagement for a Social Robot*, IEEE, Istanbul, Turkey, 2006.

[20] Sugar Sweet Robotics, "python_pepper_kinematics: Kinematics library for the pepper robot," https://github.com/sugarsweetrobotics/python_pepper_kinematics, n.d., accessed: 28-Jan-2026.

[21] Z. Lin, K. Wu, R. Shen, X. Yu, and S. Huang, "An efficient and accurate a-star algorithm for autonomous vehicle path planning," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 6, pp. 9003–9008, 2024.

[22] DFRobot, "Rplidar a1m8 - 360 degree laser scanner development kit," https://www.dfrobot.com/product-1461.html, 2024, accessed: 02-Feb-2026.

[23] V. Rabaud, K. Knese, and M. Arguedas, "naoqi_driver2: Ros2 driver for nao and pepper robots," GitHub Repository, Tech. Rep., 2024. [Online]. Available: https://github.com/ros-naoqi/naoqi_driver2

[24] NASA, "Nasa task load index," https://humansystems.arc.nasa.gov/groups/tlx/downloads/TLXScale.pdf, 1986.